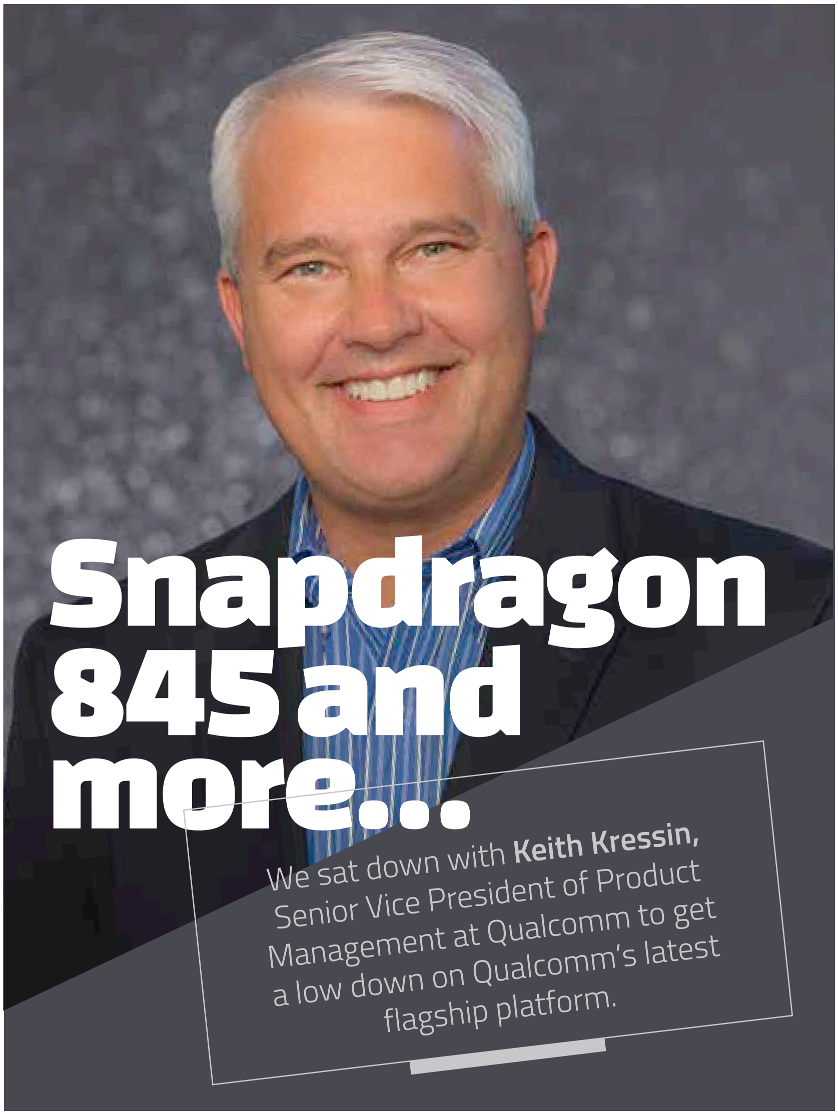

Snapdragon 845 and more...

We sat down with **Keith Kressin,** Senior Vice President of Product Management at Qualcomm to get a low down on Qualcomm's latest flagship platform.

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While the Snapdragon 845 is a flagship chip with the latest and greatest in terms of compute power, security, AI, and connectivity, how much of that will mid-tier and low-end devices get?
Kressin: We introduce a new premier tier every year and we waterfall some of the features to lower tier devices. Things like power savings and camera features will flow down to lower tiers. We will have to decide on how much of the core features such as security, image processing, etc. waterfalls down to the lower tier. Some of the flagship features might make it to lower platform after a year or two. It is a delicate balance between building the best possible platform and experience versus selecting and bringing some of the key features down to the lower tier while keeping the price points and different markets in mind. Features like XR (eXtended Reality) are compute intensive, and hence, might not be made available for lower tiers where the available processing power is limited to provide a smooth XR experience.

Samsung's flagships in India run on Samsung's Exynos processor, while they run the latest Snapdragon platform in other countries. Given the popularity of Snapdragon processors, don't you think consumers should be given the choice?
Kressin: The lead of Samsung's handset division gets to decide that. We just try to put the best products out there. How Samsung decides on using its own processors or our processors in different geographies is purely under Samsung's control. We obviously want to encourage Samsung to use our processors but for whatever reasons, sometimes they use their own.

With the Snapdragon 845, you are introducing a new level of cache - "system cache". How does it affect performance, given that this 3 MB of system cache will add another "hop", and will be the slowest of all levels of cache? Will this be used in mid-tier platforms?
Kressin: The system cache is shared across GPUs, CPUs, DSPs and other sub systems. By having the extra cache, we are looking to speed-up performance by taking some load off the RAM. Using system cache for mid-tier platforms is an option for us, but we haven't made any announcements on that so far.

When it comes to performance of devices running on Qualcomm's flagship platforms, we often see a difference in terms of speed and overall experience between two devices running on the same flagship platform. What is Qualcomm doing to solve this issue, to ensure that devices on its flagship platform perform nearly identically?
Kressin: In the end, it is up to the OEM. There's a lot that goes into the device in terms of software from us, software from the OEM and software from the platform (Google). Each OEM makes its own decisions, but in the end, one OEM has 10,000 engineers working on flagships while another has 10 engineers. Thus, the optimisation levels are different and while we help all OEMs optimise their devices, it depends on the decisions they make in terms of other components and efforts being put into software optimisation.

You mentioned that the best cameras ranked by DxOMark ran on Snapdragon's flagship platforms, with the Google Pixel 2 right at the top. However, Google uses its own visual core processor, which is an 8-core dedicated processor that does all the heavy lifting...
Kressin: Let me clarify. When the Pixel 2 was launched, and it got the best DXO Mark scores, the Visual Core was on the platform, but was turned off, and all the image processing was done by the Snapdragon 835 platform. My understanding is that, since then, Google has provided a software update and has turned on the Visual Core. It's up to Google to decide how much processing happens on its visual core and how much on the Snapdragon chip.